

Titanic Survival Analysis

Who survived, why it matters, and what it teaches us about risk & segmentation

Growth & Insight Analytics Team

Data source: Titanic passenger manifest (n = 891)

Three questions this analysis answers

01

Which passenger segment had the highest survival rate?

Class × Gender × Age

Descriptive analysis

02

Does ticket class (Pclass) impact survival?

Business lens: premium vs. economy experience

Chi-square test

03

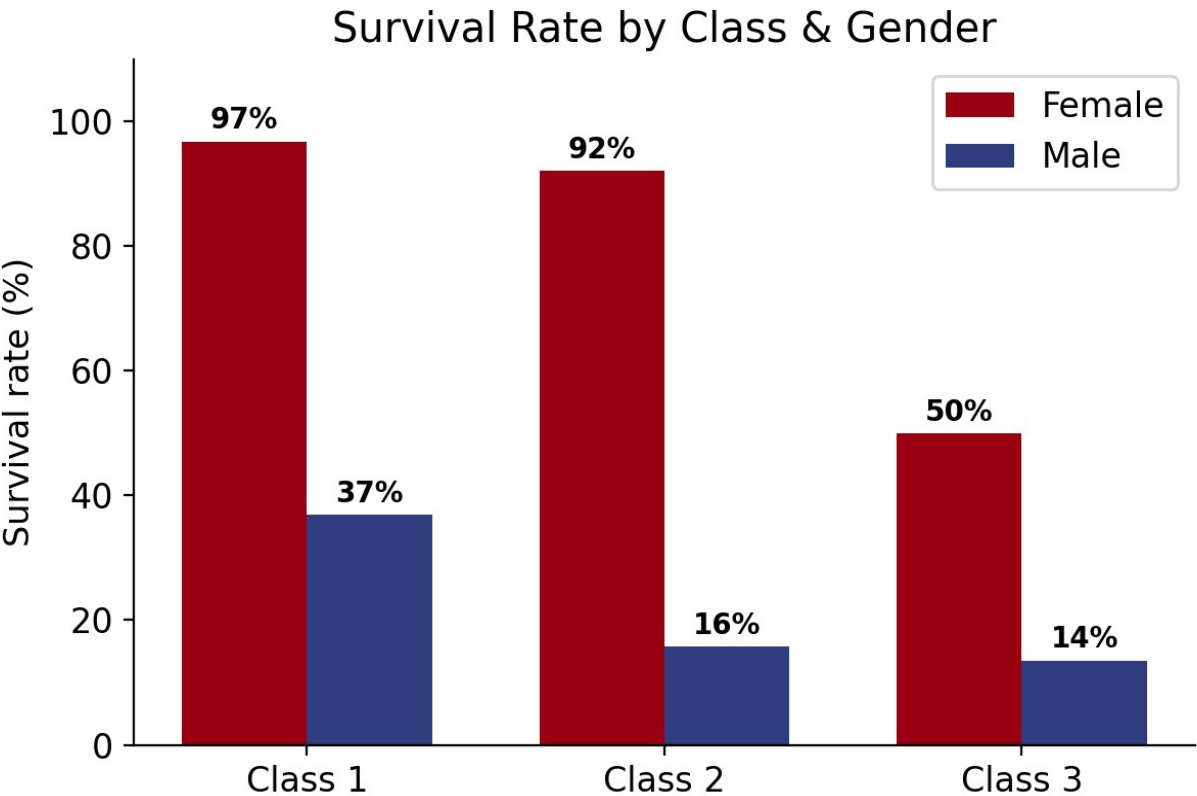
Is the survival gap between males and females statistically significant?

Hypothesis testing

Chi-square test

Method: chi-square tests of independence for categorical comparisons, tested at 95% confidence ($p < 0.05$ = statistically significant).

Q1: 1st-Class women had the highest survival rate



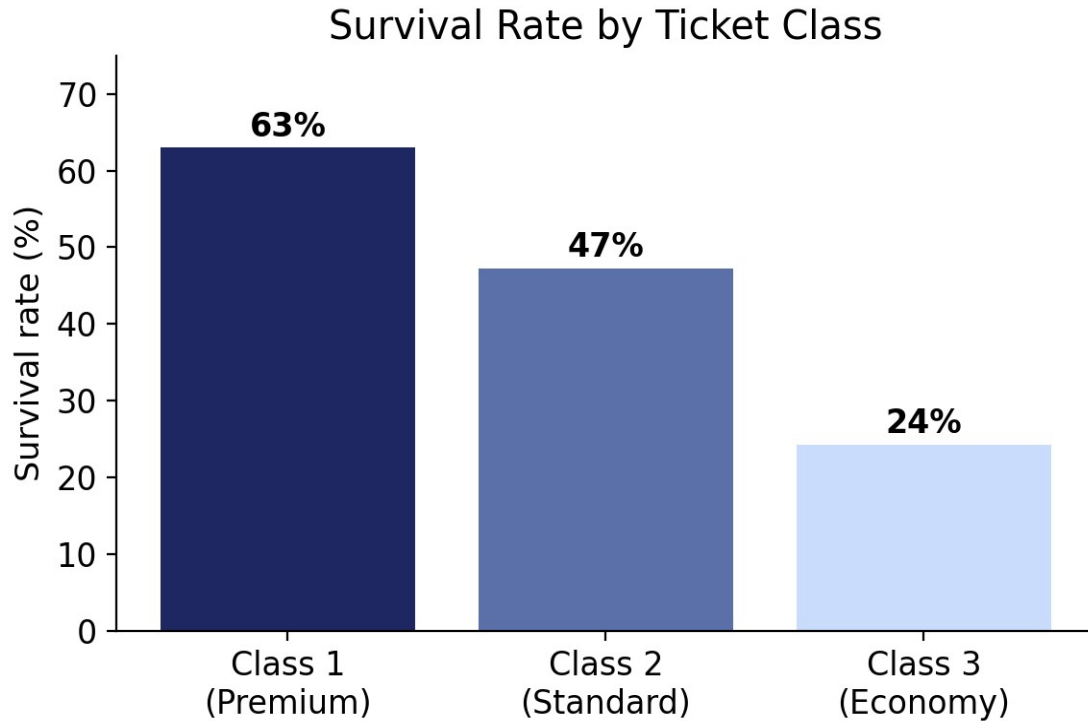
97%

Survival rate for 1st-Class women (n=94) — the highest of any Class × Gender × Age combination

The pattern holds across age bands too

1st-Class teen and adult women both survived at 95%+, while 3rd-Class men (any age) survived at under 20%. Class and gender compound each other.

Q2: Ticket class has a large, significant effect on survival



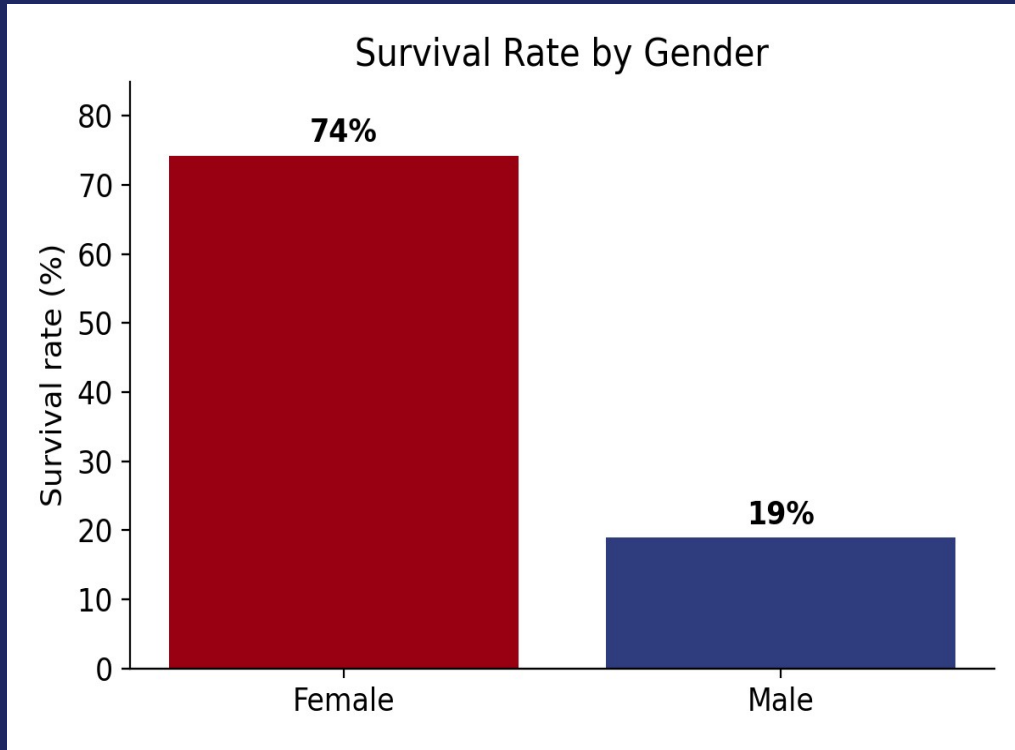
63% vs 24%

Class 1 vs. Class 3 survival rate · $\chi^2 = 102.9$, $p < 0.001$

Business insight: premium vs. economy experience

Survival climbs steadily with class — 24% → 47% → 63%. In a modern product, this mirrors how a premium tier's better access (lifeboats → priority support, faster resolution) directly changes outcomes, not just perception.

Q3: The gender gap is real, not random



74% vs 19%

Female vs. male survival · $\chi^2 = 260.7$, $p < 0.001$ ($n = 891$)

Hypothesis test verdict:

H_0 (no difference between genders) is rejected with very high confidence. The 55-point gap is the single strongest, most significant effect in the whole dataset — far too large to be explained by chance.

Recommendation

Any risk, retention, or resource-allocation model built on this kind of data must treat class and gender as independent, compounding factors — not proxies for one another — and validate that any “premium” advantage translates into a real outcome difference, not just a perceived one.